Homework 2

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February 8, 2025
MA 542 Section A1

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Let $R \coloneqq \mathbb{Z}[\sqrt{2}]$, and let $\phi : R \to R$ be a nontrivial homomorphism.

Problem 1

Note that $\phi(1) = 0$ would imply that for all $r = a + b\sqrt{2} \in R$,

$$\phi(r) = \phi(a + bi) = \phi(a) + \phi(b)\phi(\sqrt{2}) = \phi\left(\sum_{0<i\leq a} 1\right) + \phi\left(\sum_{0<j\leq b} 1\right)\phi(\sqrt{2})$$

$$= \sum_{0<i\leq a} \phi(1) + \left(\sum_{0<j\leq b} \phi(1)\right)\phi\left(\sqrt{2}\right) = 0 + 0\sqrt{2} = 0.$$

Therefore, $\phi(1) \neq 0$. Let $\phi(1) = a$. Consider the number 4. We have

$$\phi(4) = \phi(1 + 1 + 1 + 1) = \phi(1) + \phi(1) + \phi(1) + \phi(1) = 4a.$$

We also have

$$\phi(4) = \phi(2\cdot 2) = \phi(2)\phi(2) = \phi(1+1)\phi(1+1) = (\phi(1) + \phi(1))(\phi(1) + \phi(1)) = (2a)(2a) = 4a^2.$$

Since ϕ is a homomorphism, we have $4a = 4a^2$. Since $\mathbb{Z}$ is a domain, we have $a = a^2$, and since $a \neq 0$ and $\operatorname{char}\mathbb{Z} \neq 2$, it follows that $a = 1$. Therefore, $\phi(1) = 1$. Now for any $r \in \mathbb{Z}$,

$$\phi(r) = \phi\left(\sum_{0<i\leq r} 1\right) = \sum_{0<i\leq r} \phi(1) = \sum_{0<i\leq r} 1 = r.$$

Problem 2

Since ϕ is a homomorphism,

$$\phi(a + b\sqrt{2}) = \phi(a) + \phi(b\sqrt{2}) = \phi(a) + \phi(b)\phi(\sqrt{2}).$$

By part 1, $\phi(a) = a$ and $\phi(b) = b$. Therefore,

$$\phi(a) + \phi(b)\phi(\sqrt{2}) = a + b\phi(\sqrt{2}).$$

Due to this, ϕ can be viewed as a $\mathbb{Z}$-module homomorphism.

Problem 3

Let $\phi(\sqrt{2}) =: a$. Note that $a^2 = \phi(2) = 2$ by part 1. If $a = r + s\sqrt{2}$, we must have

$$\left(r + s\sqrt{2}\right)^2 = r^2 + 2s^2 + 2rs\sqrt{2} = 2.$$

Clearly, we must have $2rs\sqrt{2} = 0$,[1] and since $\mathbb{Z}$ is a domain, we must have either $r = 0$ or $s = 0$. If $s = 0$, then $r^2 = 2$, and $r = \sqrt{2} \notin \mathbb{Z}$, a contradiction. Therefore, $r = 0$. It follows that $2s^2 = 2$, and thus $s = \pm 1$. Therefore,

$$\phi(\sqrt{2}) = 1\sqrt{2} \text{ or } \phi(\sqrt{2}) = -1\sqrt{2}.$$

[1] I don't feel like proving that $a + b\sqrt{2} = c + d\sqrt{2}$ iff $a = c$ and $b = d$. This can be shown by proving $\{1, \sqrt{2}\}$ is a basis for R when viewed as a $\mathbb{Z}$-module, then applying the fact that basis representations are unique but that's like so much work for something so obvious